

Customer Lifetime Value (CLV) & Churn Prediction

Innovexa Catalyst Data Science Case Study

Author: Solasa Sahithi Vasavi

Project Objective

Develop machine learning models to predict Customer Lifetime Value (CLV) and Customer Churn using customer demographics, transactions, engagement, and support data.

Dataset

Six datasets were analyzed: customers, transactions, events, support tickets, churn labels, and CLV labels.

Methodology

Performed data cleaning, exploratory data analysis, feature engineering (RFM, engagement, support features), and trained Random Forest Regression and Classification models.

Results

CLV Model: $R^2 = 94.90\%$, MAE = 41.64, RMSE = 52.07. Churn Model: Accuracy = 89.40%, Precision = 72.73%, Recall = 47.68%, F1 = 57.60%, ROC-AUC = 72.25%.

Key Insights

Monetary Value is the strongest CLV predictor, while Refund Count is the most influential factor for churn prediction.

Business Recommendations

Increase customer spending through personalized offers, encourage premium plan upgrades, proactively monitor refunds, and target high-risk customers with retention campaigns.

Model	Best Metric
CLV Prediction	$R^2 = 94.90\%$
Churn Prediction	Accuracy = 89.40%